

December 06, 2024

Project Group #20

High-Fidelity Hearing Protection Device to Improve Sensitivity for Patients with Hyperacusis & Tinnitus

Aly Khan Nuruddin

Albin Soni

Andre Yu

Jai Choraria

Pierce Alikusumah

Table of Contents

- 01. Problem Description
- 02. User Profile
- 03. Empathy Development Plan
- 04. Needs Statements
- 05. Design Requirements
- 06. Evaluation Criteria
- 07. Concept Generation
- 08. Client Feedback
- 09. References

Problem Description

- Tinnitus (ringing in the ears) affects 10-20% of the worldwide population [1]
- Worst symptoms occurring in 10% of patients
 - Interferes with patient's ability to perform normal tasks [2]
- Often caused by acoustic trauma due to loud noise exposure.
- Often accompanied by hyperacusis (sound sensitivity) [3]
 - Many new patients will overuse hearing protection
 - Can be distressing to hear ringing and require earplugs all the time
- Need solution to help new tinnitus/hyperacusis patients acclimate to condition

User Profile

- Total Accessible Market: All individuals who have tinnitus in the world
 - Estimated to be around 750 million people in 2022 [4]
- To narrow the scope we focus on: individuals who are unsatisfied with current solutions to manage tinnitus



User Profile



Team Member:
Andre Yu

- Suffers from tinnitus and hyperacusis
 - Loud noise exposure
- Primary stakeholder
- Extensive knowledge of common problems experienced by new tinnitus/hyperacusis patients

Empathy Development Plan

1. Independent research on tinnitus → resources provided by Andre
2. Followed up with regular team meetings to share and reflect on findings. Facilitated the identification of:
 - a. Key terminologies
 - b. Main issues with existing tinnitus management
 - c. Empathy driven communication
3. Helped inform respectful and appropriate communication with end user

Needs Statements

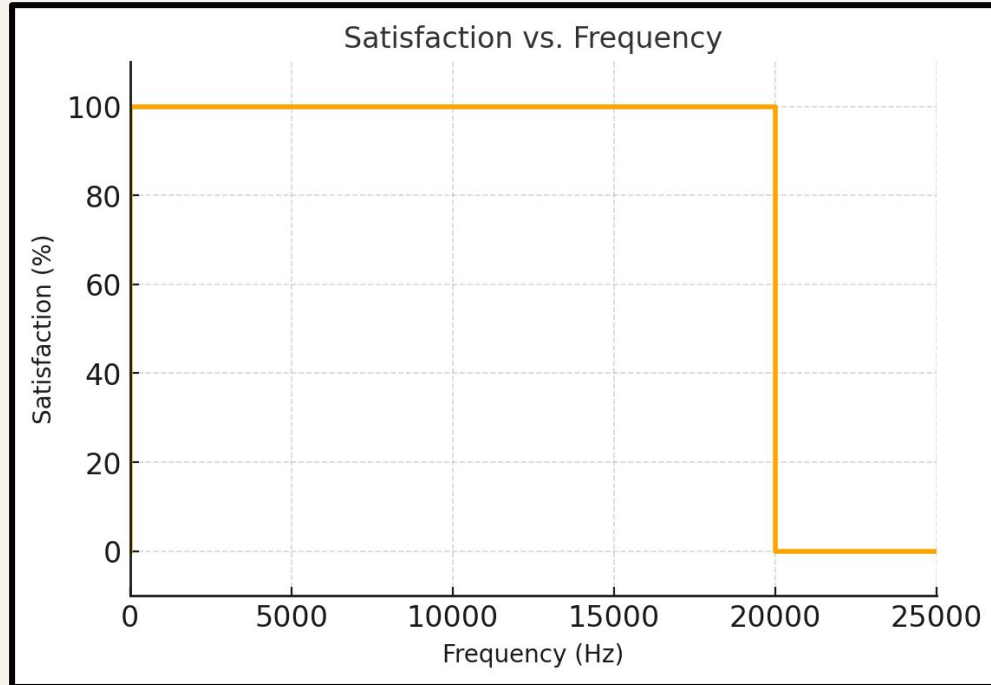
No.	Needs Statement	Classification
1.	<i>Must be able to detect between various acoustic frequencies</i>	Expressed Needs
2.	<i>Must be resistant to small changes in air pressure</i>	
3.	<i>Must detect sudden changes in the intensity of sound</i>	
4.	<i>Must not use any form of noise-cancellation</i>	
5.	<i>Must be cost-effective and cheaper than existing solutions</i>	Threshold Needs
6.	<i>Must fit snugly and discreetly on the ear lobe for regular use</i>	
7.	<i>Must be biocompatible and adhere to regulatory standards</i>	Latent Need

Design Requirements

No.	Requirement	Need
1.	<i>Must be able to detect all frequencies within the range of human hearing (20-20,000Hz)</i>	Data Quality & Accuracy
2.	<i>Must be able to detect at least ~20 dB of immediate volume change</i>	
3.	<i>Must have a Noise Reduction Rating (NRR) of at least ~25 dB</i>	
4.	<i>Must not weigh more than 4 g</i>	Ergonomics & User Compatibility
5.	<i>Must not be less than 6-8mm in length so as to not extend beyond the tragus of the ear</i>	
6.	<i>Must have a maximum market price of \$70 per unit</i>	Affordability

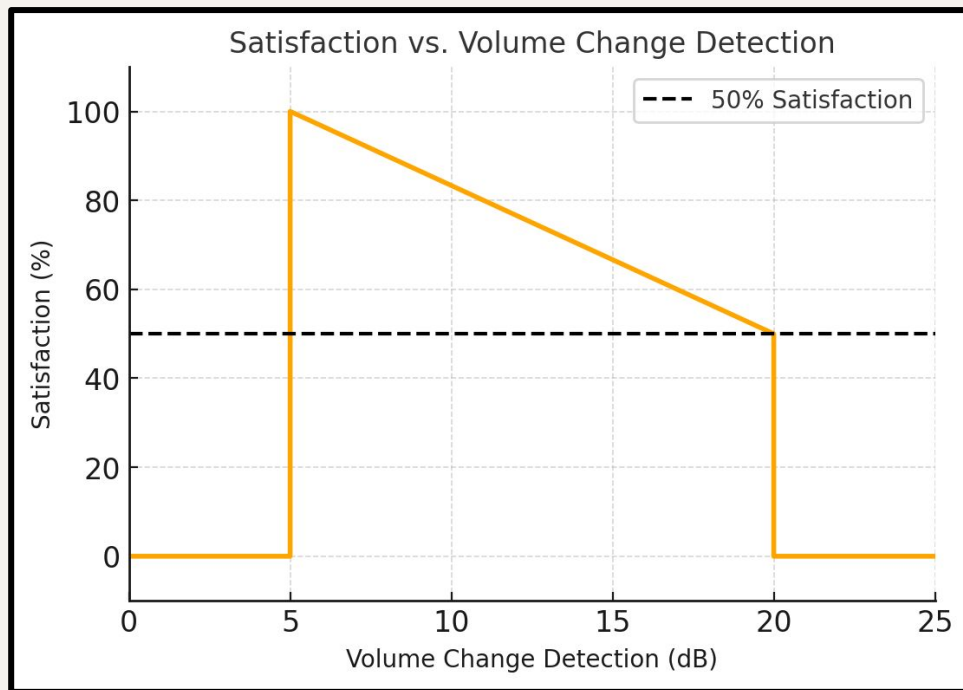
Evaluation Criteria

Criterion #1: Frequency-Detection (5)



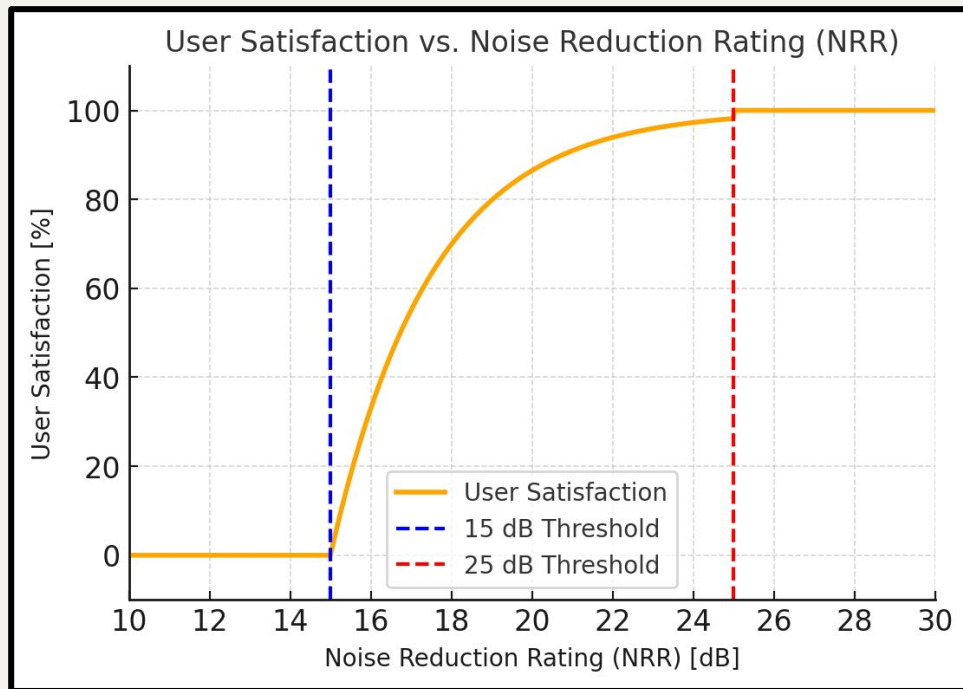
Evaluation Criteria

Criterion #2: Volume-Change Detection (6)



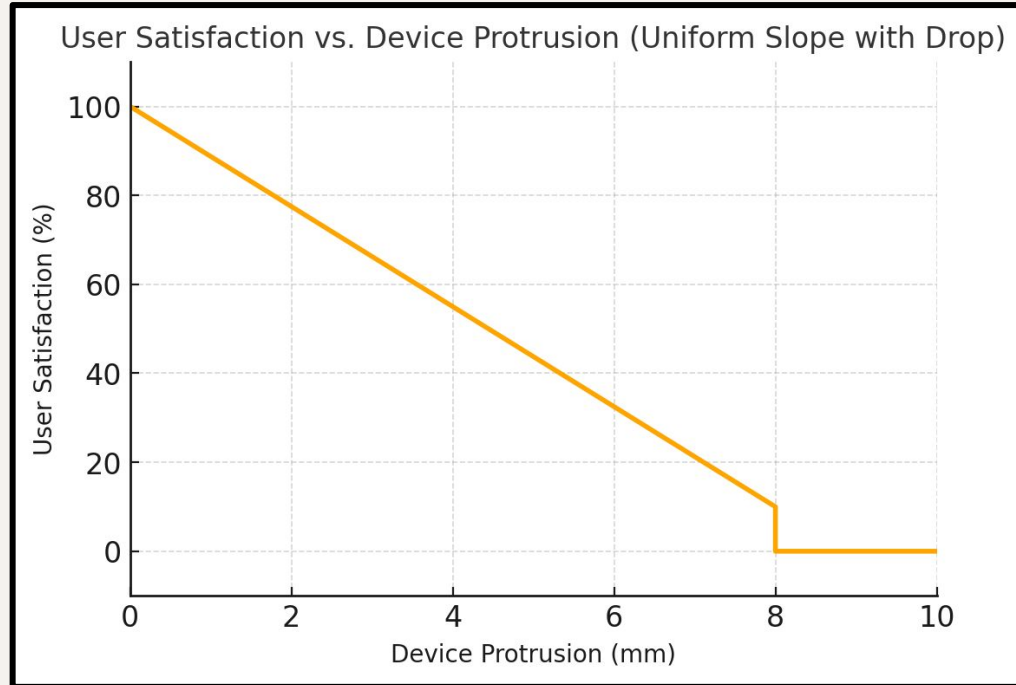
Evaluation Criteria

Criterion #3: Noise-Reduction Performance (7)



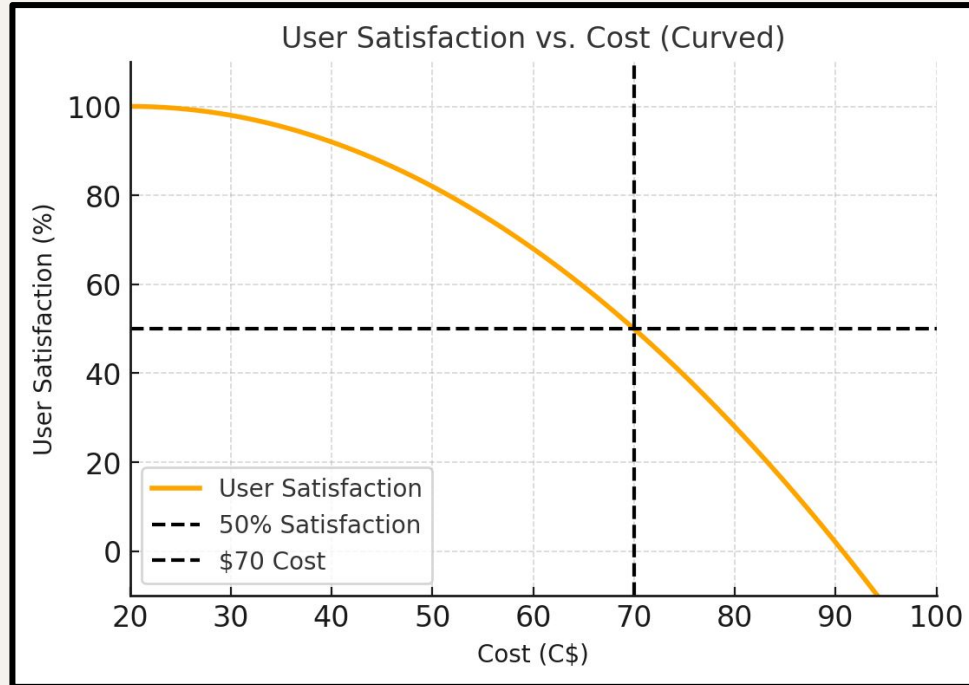
Evaluation Criteria

Criterion #4: Device Protrusion (8)



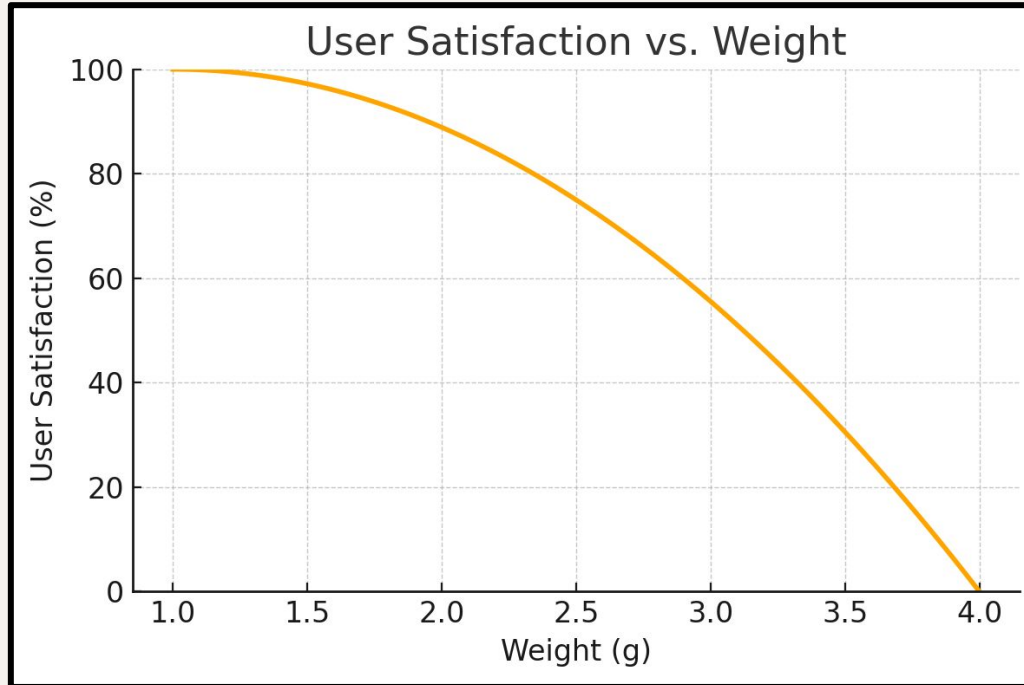
Evaluation Criteria

Criterion #5: Market Cost (9)



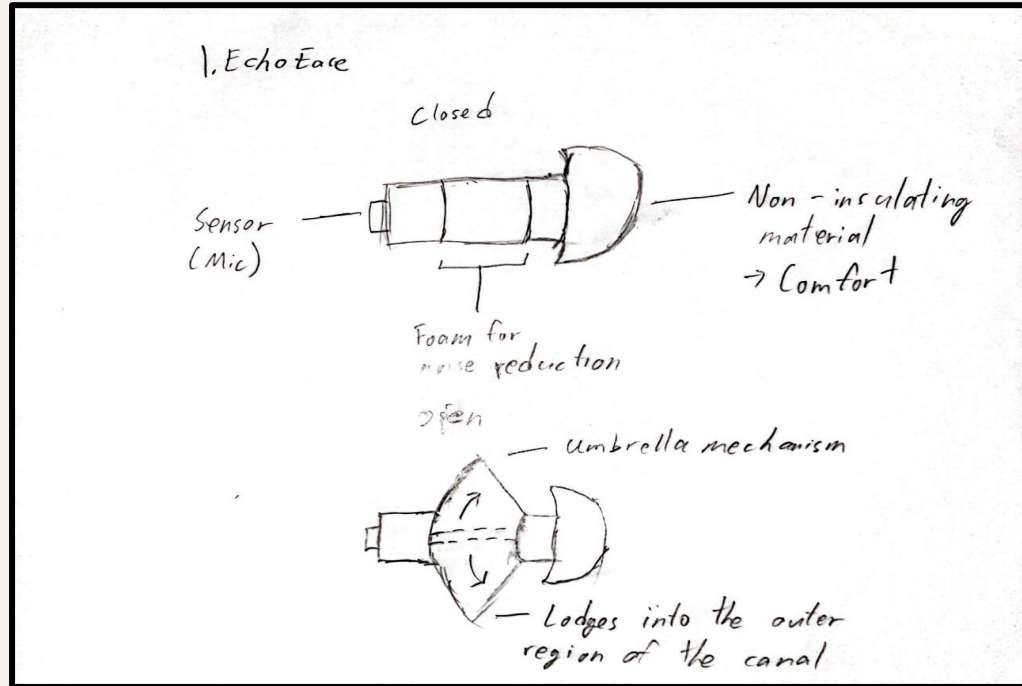
Evaluation Criteria

Criterion #6: Weight Constraints (10)



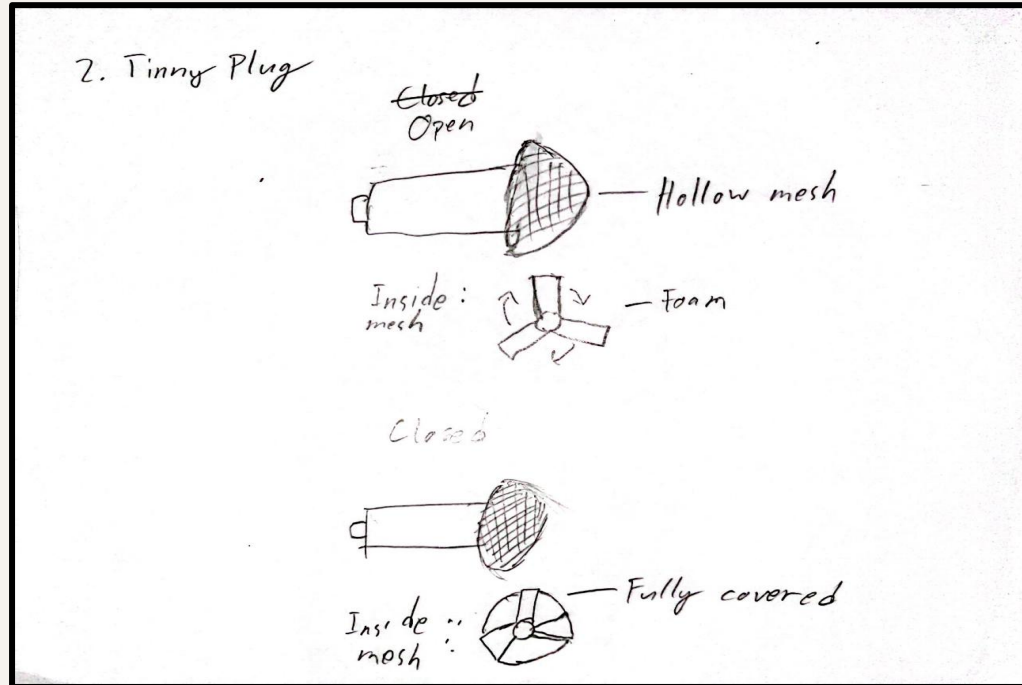
Concept Generation

Concept #1: Echo Ease



Concept Generation

Concept #2: Tinny Plug



Client Feedback

- **Concept #1**
 - Pros:
 - Overall a promising concept
 - Allows for sound to pass through with mesh design
 - Deep in-ear design is favourable
 - Cons:
 - Concerns about the noise generated when the device deploys
 - Retraction mechanism could cause loud vibrations
- **Concept #2**
 - Pros:
 - Mesh design to allow for sound passage is a good addition
 - The deployment mechanism is favourable
 - Deep in-ear design is favourable
 - Cons:
 - Concerns about whether the retraction mechanism will have sufficient NRR to adequately deploy when needed

Thank You

Questions?

References

1. Tinnitus [Internet]. NIDCD. [cited 2024 Nov 29]. Available from: <https://www.nidcd.nih.gov/health/tinnitus>
2. Tuley D. Everything you need to know about VA disability for tinnitus [Internet]. The Tuley Law Office. Tuley Law; 2024 [cited 2024 Nov 29]. Available from: <https://www.tuleylaw.com/veterans-disability-law/blog/everything-you-need-to-know-about-va-disability-for-tinnitus/>
3. About ATA [Internet]. American Tinnitus Association | The ATA is a non-profit that empowers tinnitus patients, health-care providers, and researchers. American Tinnitus Association; 2020 [cited 2024 Nov 29]. Available from: <https://www.ata.org/about-ata/>
4. Jarach CM, Lugo A, Scala M, van den Brandt PA, Cederroth CR, Odone A, et al. Global prevalence and incidence of tinnitus: A systematic review and meta-analysis. JAMA Neurol [Internet]. 2022 [cited 2024 Nov 29];79(9):888. Available from: <https://jamanetwork.com/journals/jamaneurology/fullarticle/2795168>
5. **San Francisco Audiology.** What is considered a normal hearing range? [Internet]. San Francisco Audiology. [cited 2024 Nov 29]. Available from: <https://sfaudiology.com/what-is-considered-a-normal-hearing-range/>
6. Moore BCJ, Glasberg BR, Stone MA. Development and evaluation of a method for assessing the effectiveness of hearing protectors against environmental noise. Int J Audiol [Internet]. 2021 Nov [cited 2024 Nov 29];60(11):802-811. Available from: <https://pmc.ncbi.nlm.nih.gov/articles/PMC8583416/>
7. MusicRadar. Best earplugs for musicians: Protect your hearing without losing sound quality [Internet]. MusicRadar. 2023 [cited 2024 Nov 29]. Available from: <https://www.musicradar.com/news/best-earplugs-for-musicians>
8. OHS Online. Ear plug selection and fitting best practices [Internet]. Occupational Health & Safety Online. 2017 Mar [cited 2024 Nov 29]. Available from: <https://ohsonline.com/articles/2017/03/01/ear-plug-selection-and-fitting-best-practices.aspx>
9. **My Safe Hearing.** Best hearing protection for concerts [Internet]. My Safe Hearing. 2023 [cited 2024 Nov 29]. Available from: <https://mysafehearing.com/best-hearing-protection-for-concerts/>
10. Gunaf A. Best earbuds for tinnitus relief: Top picks for a quieter, more peaceful life [Internet]. LinkedIn Pulse. 2024 [cited 2024 Nov 29]. Available from: <https://www.linkedin.com/pulse/best-earbuds-tinnitus-relief-top-picks-quieter-more-peaceful-abbas-gunaf>